

A Conditional Lift of Convex Normal Tilings to Schauder-Basis Spaces

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Status

This is a conditional reduction packet. It proves a complete implication under an explicitly stated unproved finite-dimensional hypothesis. It does not solve the open problem of Deville–Jimenez-Sevilla in full.

Source and Open Problem

The source paper is:

Robert Deville and Mar Jimenez-Sevilla, *Normal and starlike tilings in separable Banach spaces*, arXiv:2008.10367.

In the introduction, after proving universal normal starlike tilings in separable Banach spaces, the authors state the following open problem.

(separable) space nor on the norm. Also it is worth to mention that these normal and starlike tilings can be defined in such a way that some of the tiles are convex.

Recall that the Banach-Mazur distance from any n -dimensional space X to ℓ_2^n satisfies $\text{dist}(X, \ell_2^n) \leq \sqrt{n}$. Since ℓ_2^n has a 2-normal convex tiling, then X has a $2\sqrt{n}$ -normal convex tiling. The question whether every separable Banach space X (finite or infinite dimensional) has a K -normal convex tiling with constant of normality K independent of the space and the norm remains open. We do not know whether a positive answer to the finite dimensional case provides a positive answer to the infinite dimensional case.

2. NORMAL AND STARLIKE TILINGS IN SEPARABLE BANACH SPACES

Transcription of the relevant part: the question whether every separable Banach space, finite or infinite dimensional, has a K -normal convex tiling with normality constant K independent of the space and norm remains open. The authors also say that they do not know whether a positive answer in finite dimensions implies a positive answer in infinite dimensions.

Definitions

A tiling $\{T_i\}$ of a Banach space X is K -normal if there are $r > 0$ and points $x_i \in T_i$ such that

$$B(x_i, r) \subset T_i \subset B(x_i, Kr)$$

for all i . It is convex if every tile is convex.

For a finite-dimensional Banach space V , write $\tau(V)$ for the infimum of all constants S such that V admits an S -normal convex tiling.

Conditional Dependencies

The main theorem below has a local hypothesis:

$$\sup_k \tau(V_k) < \infty$$

for the initial finite-dimensional spans of a Schauder basis. The source open problem's finite-dimensional part would imply this hypothesis with a uniform constant for every basis.

No proof of the finite-dimensional hypothesis is supplied here.

Idea of the Proof

Deville–Jimenez-Sevilla build convex “cylinders” C_j^k around the finite-dimensional spaces V_k , then refine every cylinder by a starlike tiling of V_k . Their Proposition 2.6 observes that for any fixed finite number of layers, one may instead slice those layers by convex tilings of V_k ; the normality constant then depends on the largest projection norm and largest finite-dimensional tiling radius used among those layers.

If the basis projections are uniformly bounded and the finite-dimensional convex tiling constants of all V_k are uniformly bounded, the same refinement can be made in every layer at once. Convexity is preserved because each refined tile is an intersection of a convex cylinder with the inverse image of a convex finite-dimensional tile.

Theorem 1. *Let X be an infinite-dimensional Banach space with a Schauder basis (e_k) , partial-sum projections $P_k : X \rightarrow V_k = \text{span}\{e_1, \dots, e_k\}$, and*

$$B = \sup_{k \geq 1} \|P_k\| < \infty.$$

Assume that $S = \sup_{k \geq 1} \tau(V_k) < \infty$. Then X admits a convex normal tiling.

More precisely, let a, b, r, δ be the constants from Lemma 2.1 of Deville–Jimenez-Sevilla and put

$$R = a + 2b + \frac{8}{\delta}.$$

Then X has a convex K -normal tiling for every

$$K > \frac{B}{r} (R(1 + B) + Sr).$$

Proof. We use the notation and construction of Deville–Jimenez-Sevilla. Their Step 3 constructs a tiling $\{C_j^k\}$ of X by convex cylinders, indexed with $k > 0 \Rightarrow j > 0$, and points x_j^k such that

$$B(x_j^k + V_k, r) \subset C_j^k \subset B(x_j^k + V_k, R).$$

The sets C_j^k are convex because they are intersections of inverse images of convex quotient tiles under linear quotient maps.

Since (e_k) is a Schauder basis, $X = V_k \oplus \ker P_k$. For each cylinder we choose the representative x_j^k in $\ker P_k$; this is possible because every coset modulo V_k has a unique representative in $\ker P_k$.

For each $k \geq 1$, choose an S -normal convex tiling $\{D_i^k\}_i$ of V_k , scaled so that for suitable points $d_i^k \in D_i^k$,

$$B_{V_k}(d_i^k, r) \subset D_i^k \subset B_{V_k}(d_i^k, Sr).$$

Scaling is legitimate because normality constants are invariant under homotheties. For $k = 0$ put $D_0^0 = \{0\}$.

Define

$$\tilde{C}_{0,j}^0 = C_j^0$$

and, for $k \geq 1$,

$$\tilde{C}_{i,j}^k = P_k^{-1}(D_i^k) \cap C_j^k.$$

Each $\tilde{C}_{i,j}^k$ is closed and convex.

The family covers X : first choose the unique cylinder C_j^k containing a given point up to tile boundaries, then choose i with $P_k x \in D_i^k$. The pairwise interior-disjoint property follows similarly. If two refined tiles come from different cylinders, the source cylinder tiling already has disjoint interiors. If they share the same cylinder but have different finite-dimensional pieces, then any open ball contained in their intersection would have open image under the surjective map $P_k : X \rightarrow V_k$, contradicting that the finite-dimensional tiling $\{D_i^k\}$ has pairwise interior-disjoint tiles.

It remains to record uniform normality. For $k = 0$, the source estimates give

$$B(x_j^0, r) \subset C_j^0 \subset B(x_j^0, R).$$

For $k \geq 1$, we claim

$$B\left(x_j^k + d_i^k, \frac{r}{B}\right) \subset \tilde{C}_{i,j}^k \subset B\left(x_j^k + d_i^k, R(1+B) + Sr\right).$$

For the left inclusion, if $\|x - (x_j^k + d_i^k)\| \leq r/B$, then

$$\|P_k x - d_i^k\| \leq r$$

because $P_k x_j^k = 0$, hence $P_k x \in D_i^k$. Also $\text{dist}(x - x_j^k, V_k) \leq r/B \leq r$, so $x \in B(x_j^k + V_k, r) \subset C_j^k$. Thus $x \in \tilde{C}_{i,j}^k$.

For the right inclusion, take $x \in \tilde{C}_{i,j}^k$. Since $x \in C_j^k$, choose $v \in V_k$ with

$$\|x - x_j^k - v\| \leq R.$$

Finite-dimensional subspaces are proximal, so such a nearest point may be chosen. Because $x_j^k \in \ker P_k$,

$$\|v - P_k x\| = \|P_k(v + x_j^k - x)\| \leq BR.$$

Since $P_k x \in D_i^k \subset B_{V_k}(d_i^k, Sr)$, we get

$$\begin{aligned} \|x - (x_j^k + d_i^k)\| &\leq \|x - x_j^k - v\| + \|v - P_k x\| + \|P_k x - d_i^k\| \\ &\leq R + BR + Sr = R(1+B) + Sr. \end{aligned}$$

Combining the inner and outer estimates gives the stated K -normality. $\square$

Corollary 1. *If there is a universal constant S such that every finite-dimensional Banach space has an S -normal convex tiling, then every Banach space with a monotone Schauder basis has a convex K -normal tiling with K depending only on S and on the constants a, b, r, δ from the source construction.*

Proof. For a monotone basis, $B = \sup_k \|P_k\| = 1$. The finite-dimensional hypothesis gives $\tau(V_k) \leq S$ for every k , so the theorem applies with a constant independent of the particular space and norm. $\square$

Limitations and Novelty Check

This packet does not settle the finite-dimensional convex-normal tiling problem. It also does not prove the universal infinite-dimensional statement for arbitrary separable Banach spaces. For non-monotone bases the constant obtained above depends on the basis projection constant B .

The cheap run indexes were searched for arXiv:2008.10367 and the key phrases “normal and starlike tilings”, “normal tilings”, “starlike tilings”, “convex normal tiling”, and “Voronoi”. No existing packet or attempt for this target was found. No external web novelty search was performed for this conditional reduction.

References

References

- [1] R. Deville and M. Jimenez-Sevilla, *Normal and starlike tilings in separable Banach spaces*, arXiv:2008.10367.